

Task 1: Operators

1. Create the following methods inside the Operators class:
 - Add(int a, int b) — prints the sum of two numbers
 - Subtract(int a, int b) — prints the difference
 - Multiply(int a, int b) — prints the product
 - Divide(int a, int b) — prints the result of division
2. Create another method inside the same class:
 - OddEvenFinder(int number) — determines whether the given number is Odd or Even using the Ternary Operator (? :) and prints the message "Even Number" or "Odd Number" .
3. In Program.cs:
 - Call all the methods of the Operators class.

```
3  class Operators
4  {
5      public void add(int a, int b){
6          Console.WriteLine($"Sum: {a + b}");
7      }
8
9      public void subtract(int a, int b){
10         Console.WriteLine($"Difference: {a - b}");
11     }
12
13     public void multiply(int a, int b){
14         Console.WriteLine($"Product: {a * b}");
15     }
16
17     public void divide(int a, int b){
18         if (b !=0)
19             Console.WriteLine($"Quotient: {(double) a/b}");
20         else
21             Console.WriteLine("Number is not divisiable by zero");
22     }
23
24     public void OddEvenFinder(int number)
25     {
26         string result = (number % 2 == 0) ? "even number" : "odd number";
27         Console.WriteLine(result);
28     }
```

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```
1  using System;
2
3  class Program{
4
5      static void Main(){
6
7          Operators operators = new Operators();
8
9          operators.add(10,5);
10         operators.subtract(20,5);
11         operators.multiply(5,5);
12         operators.divide(30,5);
13
14         operators.OddEvenFinder(7);
15         operators.OddEvenFinder(12);
16     }
17 }
18
19
20
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run
Sum: 15
Difference: 15
Product: 25
Quotient: 6
odd number
even number
rakeshgurung@Rakeshs-MacBook-Air workshop2 %
```

Task 2: Null Operations

1. Inside the class NullOperations, create a method named PerformNullChecks() that performs the following tasks:

2. Declare a string variable named username and assign null to it.
3. Use the Ternary Operator (? :) to check whether username is null:
 - If username is null, print: "Username is not available".
- Otherwise, print the value of username.
4. Perform the same null checking using the Null-Coalescing Operator (??).
5. Use the Null-Coalescing Assignment Operator (??=) to assign a value to username only if it is currently null.
- After assignment, print the updated value of username.
6. Finally, call the PerformNullChecks() method from Program.cs.

```
NullOperation.cs
1  using System;
2
3  class NullOperation
4  {
5      public void PerformNullChecks()
6      {
7          string? username = null;
8          string result = (username == null) ? "Username is not available" : username;
9          Console.WriteLine(result);
10
11          Console.WriteLine(username ?? "Username is not available (using ??)");
12
13          username ??= "RamKrishna";
14
15          Console.WriteLine("Updated Username: " + username);
16      }
17  }
```

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C# Program.cs

```
1  using System;
2
3  class Program{
4
5      static void Main(){
6
7          NullOperation n = new NullOperation();
8          n.PerformNullChecks();
9      }
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

- rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run
Username is not available
Username is not available (using ??)
Updated Username: RamKrishna
- rakeshgurung@Rakeshs-MacBook-Air workshop2 %

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Task 3: If Else and Switch

1. Ask user to input age and using if – else – if display:

```
C# Program.cs
1  using System;
2
3  class Program
4  {
5      static void Main()
6      {
7
8          Console.Write("Enter your age: ");
9          int age = int.Parse(Console.ReadLine());
10
11         if (age < 13)
12         {
13             Console.WriteLine("Child");
14         }
15         else if (age >= 13 && age <= 19)
16         {
17             Console.WriteLine("Teenager");
18         }
19         else
20         {
21             Console.WriteLine("Adult");
22         }
23     }
24 }
25
26
```

PROBLEMS OUTPUT TERMINAL ...  zsh     ... 

- rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run
/Users/rakeshgurung/Documents/3rd Year/AD/workshop2/Program.cs(9,29):
Warning CS8604: Possible null reference argument for parameter 's' in
int.Parse(string s)'.
Enter your age: 20
Adult
- rakeshgurung@Rakeshs-MacBook-Air workshop2 %

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Task 4: If else and Switch

1. Ask user to input a number (1–7) and using switch, print the day of the week:

```
1  using System;
2
3  class Program
4  {
5      static void Main()
6      {
7          Console.Write("Enter a number (1-7): ");
8          string input = Console.ReadLine();
9
10         int dayNumber;
11         bool isValid = int.TryParse(input, out dayNumber);
12
13         if (!isValid)
14         {
15             Console.WriteLine("Invalid input! Please enter a number.");
16             return;
17         }
18
19         // Use switch to print the day
20         switch (dayNumber)
21         {
22             case 1:
23                 Console.WriteLine("Sunday");
24                 break;
25             case 2:
26                 Console.WriteLine("Monday");
27                 break;
28             case 3:
29                 Console.WriteLine("Tuesday");
30                 break;
31             case 4:
32                 Console.WriteLine("Wednesday");
33                 break;
```

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```
Program.cs
34
35     case 5:
36         Console.WriteLine("Thursday");
37         break;
38     case 6:
39         Console.WriteLine("Friday");
40         break;
41     case 7:
42         Console.WriteLine("Saturday");
43         break;
44     default:
45         Console.WriteLine("Invalid day number!");
46         break;
47     }
48 }
49
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

zsh + v

- rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run
/Users/rakeshgurung/Documents/3rd Year/AD/workshop2/Program.cs(8,24): warning CS8600: Converting null literal or possible null value to non-nullable type.
Enter a number (1-7): 5
Thursday
- rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run
/Users/rakeshgurung/Documents/3rd Year/AD/workshop2/Program.cs(8,24): warning CS8600: Converting null literal or possible null value to non-nullable type.
Enter a number (1-7): 9
Invalid day number!
- rakeshgurung@Rakeshs-MacBook-Air workshop2 %

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Task 5: Loops

1. Ask the user to input a number N.
Using a for loop, calculate and print the sum from 1 to N.
 - Example: Input: 5 → Output: 15
2. Print numbers from 1 to 20 using while loop:
 - Skip (continue) multiples of 4
 - Stop (break) when number is 15
3. Write a program to find the sum of all elements in an array using a foreach loop.

```
C# Program.cs
1  using System;
2
3  class Program
4  {
5      static void Main()
6      {
7          Console.Write("Enter a number N: ");
8          int N = int.Parse(Console.ReadLine());
9
10         int sum = 0;
11         for (int i = 1; i <= N; i++)
12         {
13             sum += i;
14         }
15
16         Console.WriteLine($"The sum from 1 to {N} is {sum}");
17     }
18 }
19
```

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```
Enter a number (1-7): 5
● rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run
/Users/rakeshgurung/Documents/3rd Year/AD/workshop2/Program.cs(8,27): warning CS8604: Pos
sible null reference argument for parameter 's' in 'int int.Parse(string s)'.
Enter a number N: 7
The sum from 1 to 7 is 28
○ rakeshgurung@Rakeshs-MacBook-Air workshop2 %
```


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C# Program.cs

```
1  using System;
2
3  class Program
4  {
5      static void Main()
6      {
7          int i = 1;
8
9          while (i <= 20)
10         {
11             if (i % 4 == 0)
12             {
13                 i++;
14                 continue; // Skip multiples of 4
15             }
16
17             if (i == 15)
18             {
19                 break; // Stop at 15
20             }
21
22             Console.WriteLine(i);
23             i++;
24         }
25     }
26 }
27
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

● rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run

```
1
2
3
5
6
7
9
10
11
13
14
```

○ rakeshgurung@Rakeshs-MacBook-Air workshop2 %

```
Program.cs
1  using System;
2
3  class Program
4  {
5      static void Main()
6      {
7          int[] numbers = { 2, 4, 6, 8, 10 };
8          int sum = 0;
9
10         foreach (int num in numbers)
11         {
12             sum += num;
13         }
14
15         Console.WriteLine($"The sum of all elements is {sum}");
16     }
17 }
18
```

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● rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run
The sum of all elements is 30
○ rakeshgurung@Rakeshs-MacBook-Air workshop2 %

Task 6: Try Catch Finally

1. Write a program that accepts a number from the user and converts it to an integer.
 - o If user enters invalid input (e.g., letters), catch the exception and show: "Invalid number format".
 - o Display "Program Executed" in finally block.
2. Write a program that asks for a password.
 - o If password length is less than 6 characters, use throw to raise exception: "Password must be at least 6 characters."

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- o Otherwise, print "Password satisfied the requirement".

```
C# Program.cs
1  using System;
2
3  class Program
4  {
5      static void Main()
6      {
7          Console.Write("Enter a number: ");
8          string input = Console.ReadLine();
9
10         try
11         {
12             int number = int.Parse(input);
13             Console.WriteLine($"You entered: {number}");
14         }
15         catch (FormatException)
16         {
17             Console.WriteLine("Invalid number format");
18         }
19         finally
20         {
21             Console.WriteLine("Program Executed");
22         }
23     }
}
```

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- rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run
/Users/rakeshgurung/Documents/3rd Year/AD/workshop2/Program.cs(8,24): warning CS8600: Converting null literal or possible null value to non-nullable type.
/Users/rakeshgurung/Documents/3rd Year/AD/workshop2/Program.cs(12,36): warning CS8604: Possible null reference argument for parameter 's' in 'int int.Parse(string s)'.
Enter a number: 5
You entered: 5
Program Executed
- rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run
Enter a number: qwerty
Invalid number format
Program Executed
- rakeshgurung@Rakeshs-MacBook-Air workshop2 %

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```
C# Program.cs
1  using System;
2
3  class Program
4  {
5      static void Main()
6      {
7          Console.Write("Enter your password: ");
8          string password = Console.ReadLine();
9
10         try
11         {
12             if (password.Length < 6)
13             {
14                 throw new Exception("Password must be at least 6 characters.");
15             }
16             else
17             {
18                 Console.WriteLine("Password satisfied the requirement");
19             }
20         }
21         catch (Exception ex)
22         {
23             Console.WriteLine(ex.Message);
24         }
25     }
26 }
27
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run
/Users/rakeshgurung/Documents/3rd Year/AD/workshop2/Program.cs(8,24): warning CS8600: Conversion of 'string' to 'int' is possible due to context but may be unintended.
rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run
/Users/rakeshgurung/Documents/3rd Year/AD/workshop2/Program.cs(8,27): warning CS8600: Conversion of 'string' to 'int' is possible due to context but may be unintended.
/Users/rakeshgurung/Documents/3rd Year/AD/workshop2/Program.cs(12,17): warning CS8602: Dereferencing a possibly null value.
Enter your password: poetry
Password satisfied the requirement
rakeshgurung@Rakeshs-MacBook-Air workshop2 % dotnet run
Enter your password: poem
Password must be at least 6 characters.
rakeshgurung@Rakeshs-MacBook-Air workshop2 %
```